

Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in JUnit

Calculator.java:

```
package com.exercise4;
```

```
public class Calculator {
```

```
    public int add(int a, int b) {
```

```
        return a + b;
```

```
    }
```

```
}
```

CalculatorTest.java:

```
package com.exercise4;
```

```
import static org.junit.Assert.assertEquals;
```

```
import org.junit.After;
```

```
import org.junit.Before;
```

```
import org.junit.Test;
```

```
public class CalculatorTest {
```

```
    private Calculator calculator;
```

@Before

```
public void setUp() {  
    System.out.println("Setting up test...");  
    calculator = new Calculator();  
}
```

@After

```
public void tearDown() {  
    System.out.println("Cleaning up...");  
    calculator = null;  
}
```

@Test

```
public void testAdd() {  
  
    // Arrange  
    int a = 2;  
    int b = 3;  
  
    // Act  
    int result = calculator.add(a, b);  
  
    // Assert  
    assertEquals(5, result);  
}  
}
```

Output:

```
PS D:\SD\Digital-Nurture-JavaFSE-main\local_repo\Cognizant-exercices\Week 1\JUnit\Exercise-4\exercise4> mvn test
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.exercise4.CalculatorTest
Setting up test...
Cleaning up...
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.048 s -- in com.exercise4.CalculatorTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.378 s
[INFO] Finished at: 2026-06-21T21:02:32+05:30
[INFO] -----
```